SHYAM LAL COLLEGE
(University of Delhi)

Complex Analysis

PRACTICAL FILE

Submitted to:

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Semester 6th

Course : B.Sc Mathematics Hons.

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1.	Make a Geometric plot to show that nth root of unity are equally spaced points that lie on the unit circle C1(0)= {Z:|Z|=1} and form the vertics of a regular polygon with n sides for n = 4,5,6,7,8		
2.	Find all the solutions of the equation and Represent these geometrically		
3.	Write parametric equations and make a parametric plot for an ellipse centred at the origin with horizontal major axis of 4 units and vertical minor axis of 2 units. Show the effect of rotation of this ellipse by an angle of π/6 radians and shifting of the centre (0,0) to (2,1), by making a parametric plot.		
4.	Show that the image of the open disk D1 (-1, i) = {z : | z + 1 + i | < 1} under the transformation w = f (z) .		
5.	Show that the image of the right half plane Re(z)=x>1 under the linear transformation w=(-1+𝕚)z1-2+3*𝕚 is the half plane v>u+7, where u=RE(w),etc. Plot the map. Procedure :If w=F(z)=(-1+i)z-2+3i, then z=(w+2-3i)/(-1+i)		
6.	Show that the image of the right half plane A = {z : Re (z) >= 1/2} under the mapping w = f (z) = 1/2 is the closed disk D1 = {w : | w - 1 | <= 1}.		
7.	Make a plot of the vertical lines x = a for a = -1, -1/2, 1/2,1 and the horizontal lines y = b for b = -1, -1/2, 1/2, 1.Find the plot of the grid under the mapping w = f (z) = 1/z		
8.	Find a parametrization of the polynomial path C=C1+C2+C3 from -1+i to 3-i, where C1 is the line from -1+i to -1,C2 is the line from -1 to 1+i and C3 is the line from 1+i to 3-i. Make a plot of this path.		
9.	Plot the segment 'L' joining the point A =0 to B=2+i(π/4) and give an exact calculation ∫L e^z ⅆz		
10.	Plot the Semicircle 'C' with radius 1 centered at z=2 and evaluate the contour integral upper ∫C 1/(z-2)ⅆz		
11.	Show that ∫c1zⅆz=∫c2zⅆz= 4+2i where c1 is the line-segment from -1-i to 3+i and c2 is the portion of the parabola x=y^2+2y joining -1-i to 3+i. Make plots of two contours c1 and c2 joining -1-i to 3+i.		
12.	Use ML-inequality to show that ∫1/(z^2+1)ⅆz|<= 1/2 sqrt 5 where C is the straight line segment from 2 to 2+i. While solving represent the distance from the point z to the points i and -i respectively i.e; |z+i| and |z i| on the complex plane C.		
13.	Show that |∫c 1/(2 (z)^1/2)ⅆz=1+i where (z)^1/2 is the principal branch of the square root function and C is the line segment joining 4 to 8+6i. Also plot the path of integration.		
14.	Find the laurtenz Series of a given function f (z) around a given point z , and also Find and Plot three different		

	Laurentz Series representation for the function $F (Z) = 3/2 + Z - Z^2$ involving power of z .		
15.	Locate the poles of $f[z] = 1/(5 z^4 + 26 z^2 + 5)$ and specify their order .		
16.	Locate the zeros and poles of $g(z)=\pi Cot(n\pi)/z^2$ and determine their order. Also justify that $Res(g,0)=-\pi^2/3$		
17.	Evaluate $\int Exp(2/z) ⅆz$, where C(0) denotes the circle $\{z:\|z\|=1\}$ with positive orientation		

06/02/2026

Practical-1

Make a geometric plot to show that the nth roots of unity are equally spaced points that lie on the unit circle C1(0)={z:|z|=1} and form the vertices of a regular polygon with n sides, for n=4,5,6,7,8

In[1]:=
```
Solve [z^5 == 1, z]
```

Out[1]= $\{\{z \to 1\}, \{z \to -(-1)^{1/5}\}, \{z \to (-1)^{2/5}\}, \{z \to -(-1)^{3/5}\}, \{z \to (-1)^{4/5}\}\}$

In[3]:=
```
roots = ComplexExpand[Solve[z^5 == 1, z]]
```

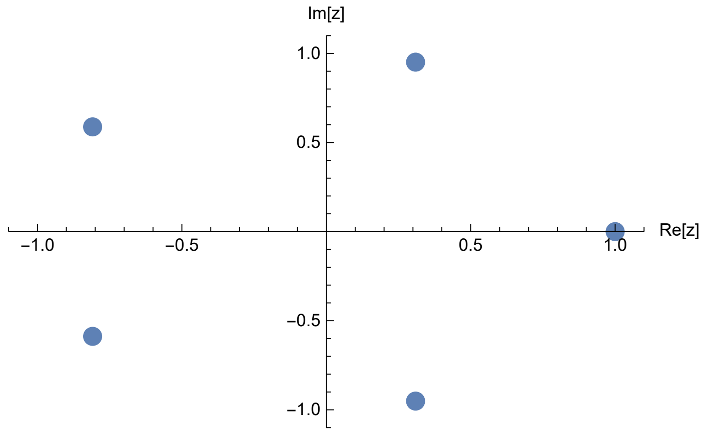
Out[3]= $$\left\{\{z \to 1\}, \left\{z \to -\frac{1}{4} - \frac{\sqrt{5}}{4} - i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}, \left\{z \to -\frac{1}{4} + \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}\right\}, \left\{z \to -\frac{1}{4} + \frac{\sqrt{5}}{4} - i\sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}\right\}, \left\{z \to -\frac{1}{4} - \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}\right\}$$

In[4]:=
```
z /. roots
```

Out[4]= $$\left\{1, -\frac{1}{4} - \frac{\sqrt{5}}{4} - i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}, -\frac{1}{4} + \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}, -\frac{1}{4} + \frac{\sqrt{5}}{4} - i\sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}, -\frac{1}{4} - \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}$$

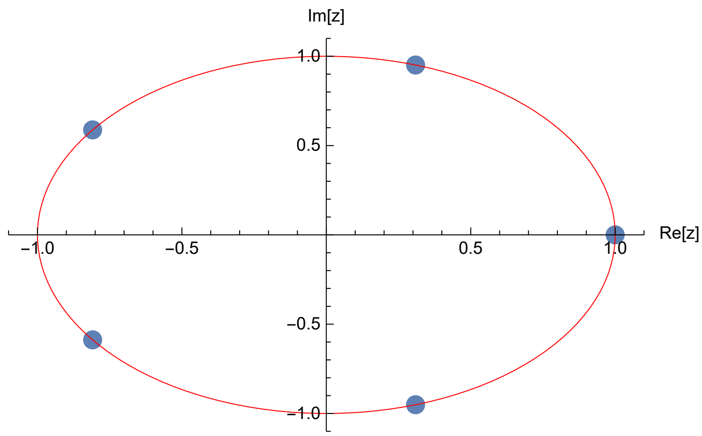
In[5]:=
```
rootPlot = ListPlot[{Re[z], Im[z]} /. roots, PlotRange → {{-1.1, 1.1}, {-1.1, 1.1}},
  AxesLabel → {"Re[z]", "Im[z]"}, PlotStyle -> PointSize[0.03]]
```

Out[5]=



In[6]:=
```
Show[rootPlot, Graphics[{Red, Circle[{0, 0}, 1]}]]
```

Out[6]=



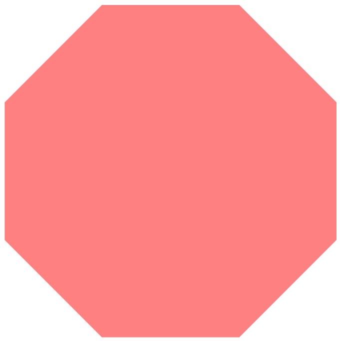
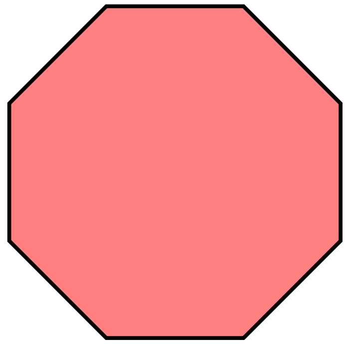
In[7]:=

$$p = \text{RegularPolygon}\left[\left\{\{1, 0\}, \left\{-\frac{1}{4} - \frac{\sqrt{5}}{4}, -\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}, \left\{-\frac{1}{4} + \frac{\sqrt{5}}{4}, \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}\right\}, \left\{-\frac{1}{4} + \frac{\sqrt{5}}{4} - \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}\right\}, \left\{-\frac{1}{4} - \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}\right\}\right]$$

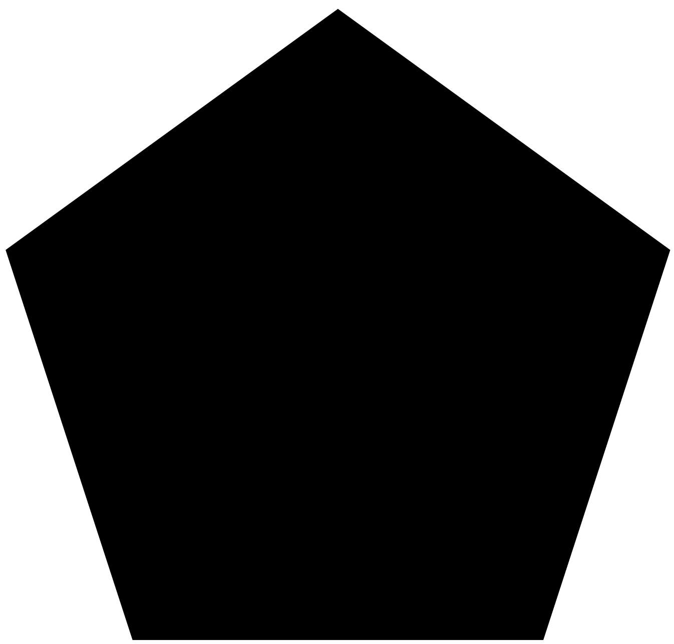
Out[7]=

$$\text{RegularPolygon}\left[\left\{\{1, 0\}, \left\{-\frac{1}{4} - \frac{\sqrt{5}}{4}, -\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}, \left\{-\frac{1}{4} + \frac{\sqrt{5}}{4}, \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}\right\}, \left\{-\frac{1}{4} + \frac{\sqrt{5}}{4} - \sqrt{\frac{5}{8} + \frac{\sqrt{5}}{8}}\right\}, \left\{-\frac{1}{4} - \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}\right\}\right\}\right]$$

```
In[10]:= R = RegularPolygon[8];
         {Graphics[{Pink, R}], Graphics[{EdgeForm[Thick], Pink, R}]}
```

Out[11]= {  ,  }

```
In[12]:= Graphics[Polygon[CirclePoints[5]]]
```

Out[12]= 

PRACTICAL -2

In[ ]:=
```
DateObject["March 13, 2026"]
```

Out[ ]= Day: **Fri 13 Mar 2026**

Find all the solutions of the equation z^3=8i and represent these geometrically.

In[ ]:=
```
Solve[z^3 == 8 * i, z]
```

Out[ ]= $\{\{z \to 2\, i^{1/3}\}, \{z \to -2\,(-1)^{1/3}\, i^{1/3}\}, \{z \to 2\,(-1)^{2/3}\, i^{1/3}\}\}$

In[ ]:=
```
roots = ComplexExpand[Solve[z^3 == 8 * i, z]]
```

Out[ ]=
$$\left\{\left\{z \to 2\left(i^2\right)^{1/6} \mathrm{Cos}\left[\frac{\mathrm{Arg}[i]}{3}\right] + 2\, \mathrm{i}\left(i^2\right)^{1/6} \mathrm{Sin}\left[\frac{\mathrm{Arg}[i]}{3}\right]\right\},\right.$$
$$\left\{z \to -\left(i^2\right)^{1/6} \mathrm{Cos}\left[\frac{\mathrm{Arg}[i]}{3}\right] + \sqrt{3}\left(i^2\right)^{1/6} \mathrm{Sin}\left[\frac{\mathrm{Arg}[i]}{3}\right] + \mathrm{i}\left(-\sqrt{3}\left(i^2\right)^{1/6} \mathrm{Cos}\left[\frac{\mathrm{Arg}[i]}{3}\right] - \left(i^2\right)^{1/6} \mathrm{Sin}\left[\frac{\mathrm{Arg}[i]}{3}\right]\right)\right\},$$
$$\left.\left\{z \to -\left(i^2\right)^{1/6} \mathrm{Cos}\left[\frac{\mathrm{Arg}[i]}{3}\right] - \sqrt{3}\left(i^2\right)^{1/6} \mathrm{Sin}\left[\frac{\mathrm{Arg}[i]}{3}\right] + \mathrm{i}\left(\sqrt{3}\left(i^2\right)^{1/6} \mathrm{Cos}\left[\frac{\mathrm{Arg}[i]}{3}\right] - \left(i^2\right)^{1/6} \mathrm{Sin}\left[\frac{\mathrm{Arg}[i]}{3}\right]\right)\right\}\right\}$$

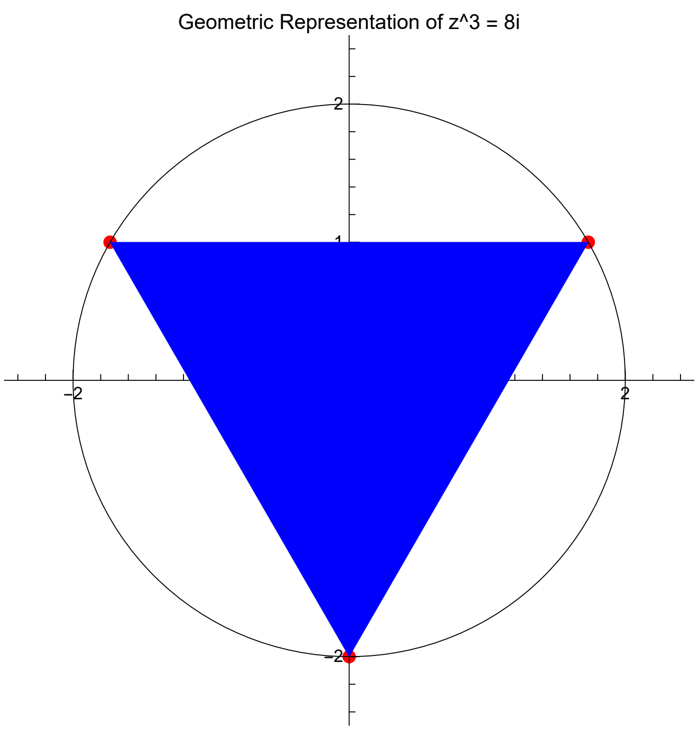
In[ ]:=
```
(*Solve the equation*)roots = z /. Solve[z^3 == 8 I, z];

(*Display numerical values*)
Print["The roots are: ", N[roots]];

(*Visualize the roots*)
ComplexListPlot[roots, PlotStyle → {PointSize[Large], Red},
 PlotRange → {{-2.5, 2.5}, {-2.5, 2.5}}, AspectRatio → 1,
 Epilog → {Circle[{0, 0}, 2], Blue, Dashed, Polygon[ReIm /@ roots]},
 PlotLabel → "Geometric Representation of z^3 = 8i"]
```

The roots are: {0. − 2. i, 1.73205 + 1. i, −1.73205 + 1. i}

Out[ ]=

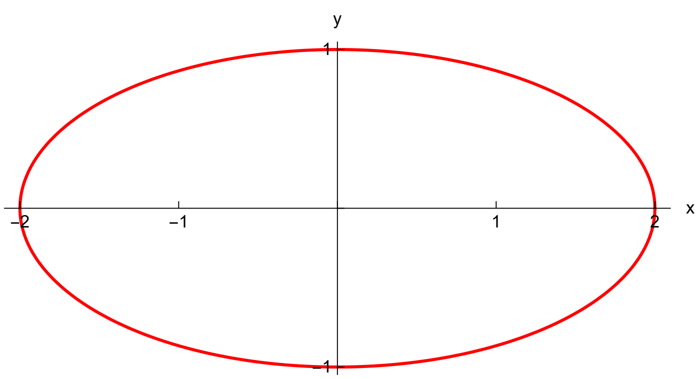


Practical -3

Write parametric equations and make a parametric plot for an ellipse centred at the origin with horizontal major axis of 4 units and verticle minor axis of 2 units. Show the effect of rotation of this ellipse by an angle of pie / 6 radians and shifting of the centre (0, 0) to (2, 1), by making a parametric plot.

In[∘]:=
```
ClearAll
```
Out[∘]=
```
ClearAll
```

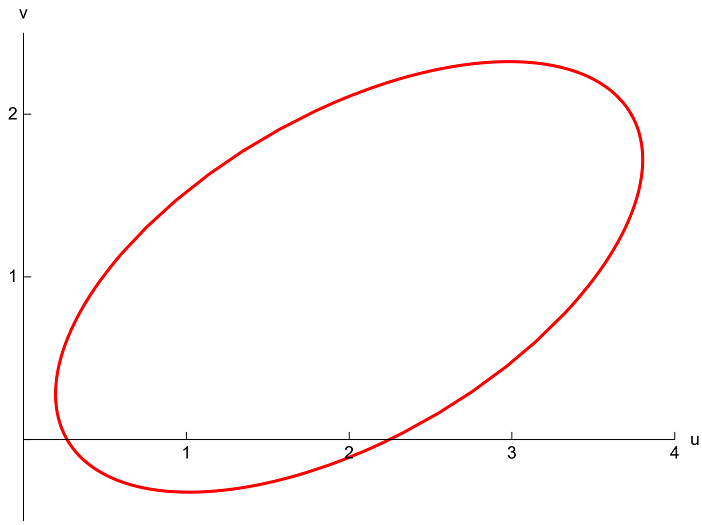
In[5]:=
```
Clear[r, s, t]
s[t_] = 2 Cos[t] + I * Sin[t];
r[t_] = s[t] * E^(I * Pi / 6) + (2 + I);
ParametricPlot[{Re[s[t]], Im[s[t]]}, {t, 0, 2 π}, PlotStyle → Red,
 PlotRange → {{-2.1, 2.1}, {-1.05, 1.05}}, AspectRatio → 1 / 2,
 Ticks → {Range[-2, 2, 1], Range[-2, 2, 1]}, AxesLabel → {"x", "y"}]
Print["s[t]=", s[t], "for 0≤t≤2π"]
```

Out[8]=



```
s[t]=2 Cos[t] + ⅈ Sin[t]for 0≤t≤2π
```

In[10]:=
```
ParametricPlot[{Re[r[t]], Im[r[t]]}, {t, 0, 2 Pi},
 PlotStyle -> Red, PlotRange -> {{0, 4}, {-0.5, 2.5}}, AspectRatio -> 3/4,
 Ticks -> {Range[0, 4, 1], Range[0, 3, 1]}, AxesLabel -> {"u", "v"}]
Print["r[t]=", r[t], "for 0≤t≤2π"]
```

Out[10]=



$r[t] = (2 + i) + e^{\frac{i\pi}{6}} (2\,\mathrm{Cos}[t] + i\,\mathrm{Sin}[t])$ for $0 \le t \le 2\pi$

Practical -4

Show that the image pf the open disk D1(-1,-i)=[z:z+1+i<1} under the linear transformation w=f(z)

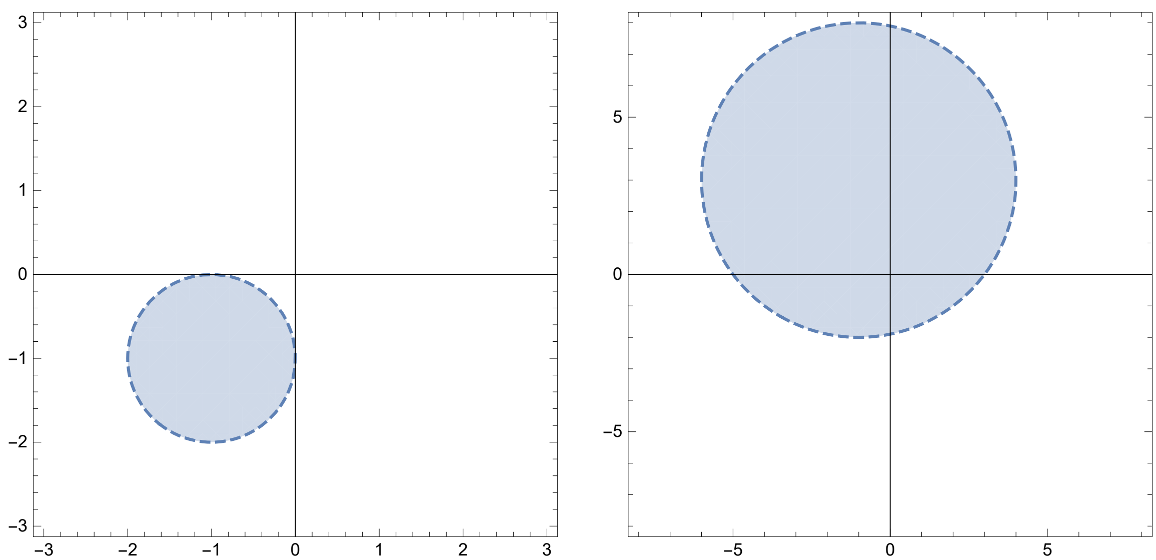
```
In[ ]:= z = x + ⅈ * y
Out[ ]= x + ⅈ y

In[ ]:= Solve[w1 == (3 - 4 * ⅈ) z1 + 6 + 2 * ⅈ, z1]
```

$$\text{Out[ ]=}\ \left\{\left\{z1 \to \frac{1}{25}\left(-10 + 3\,w1 + 2\,ⅈ\,(-15 + 2\,w1)\right)\right\}\right\}$$

```
In[ ]:= A1 = RegionPlot[Abs[z + 1 + ⅈ] < 1,
            {x, -3, 3}, {y, -3, 3}, BoundaryStyle → Dashed, Axes → True];
        A2 = RegionPlot[Abs[((z - 6 - 2 * ⅈ) / (3 - 4 * ⅈ)) + 1 + ⅈ] < 1,
            {x, -8, 8}, {y, -8, 8}, BoundaryStyle → Dashed, Axes → True];
        GraphicsRow[{A1, A2}]
```

Out[ ]=

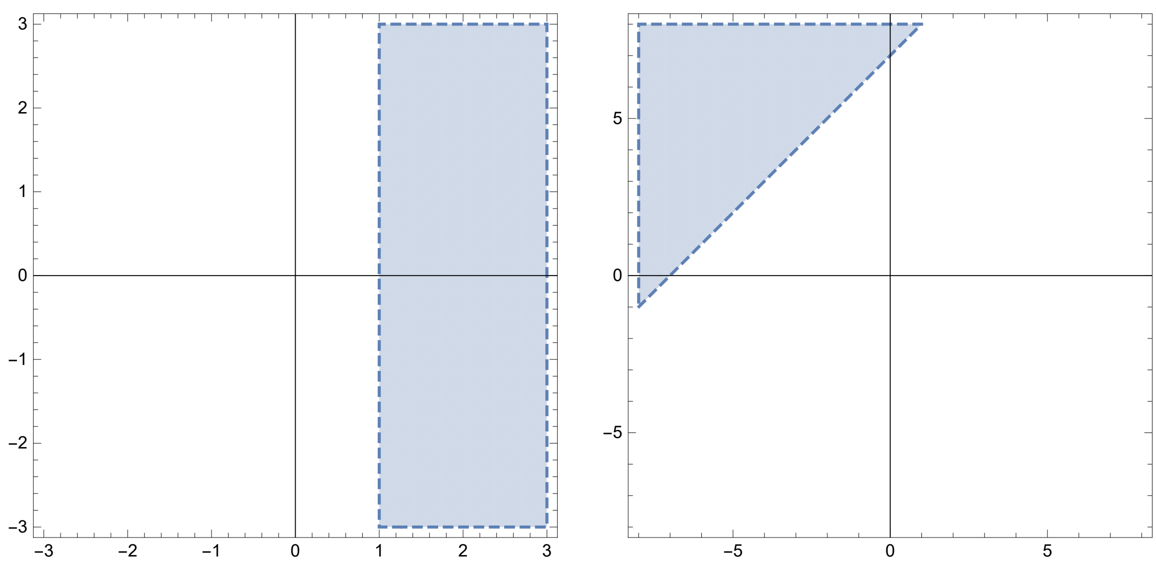


Practical-5

Show that the image of the right half plane Re(z)=x>1 under the linear transformation w=(-1+i)z1-2+3*i is the half plane v>u+7, where u=RE(w),etc.Plot the map. Procedure :If w=F(z)=(-1+i)z-2+3i, then z=(w+2-3i)/(-1+i)

In[∘]:=
```
z = x + i * y
```
Out[∘]=
```
x + i y
```
In[∘]:=
```
Solve[w1 == (-1 + i) z1 - 2 + 3 * i, z1]
```
Out[∘]= $\{\{z1 \to \frac{1}{2} (-5 + i (1 - w1) - w1)\}\}$

In[∘]:=
```
A1 = RegionPlot[Re[z] > 1, {x, -3, 3}, {y, -3, 3}, BoundaryStyle → Dashed, Axes → True];
A2 = RegionPlot[Re[(z + 2 - 3 * i) / (-1 + i)] > 1,
   {x, -8, 8}, {y, -8, 8}, BoundaryStyle → Dashed, Axes → True];
GraphicsRow[
 {A1,
  A2}]
```
Out[∘]=



Practical-06
25/03/2026

Show that the image of the right half plane A={z:Re(z)>=1/2} under the mapping w=f(z)=1/2 is the closed disk D1={w:|w-1|<=1}

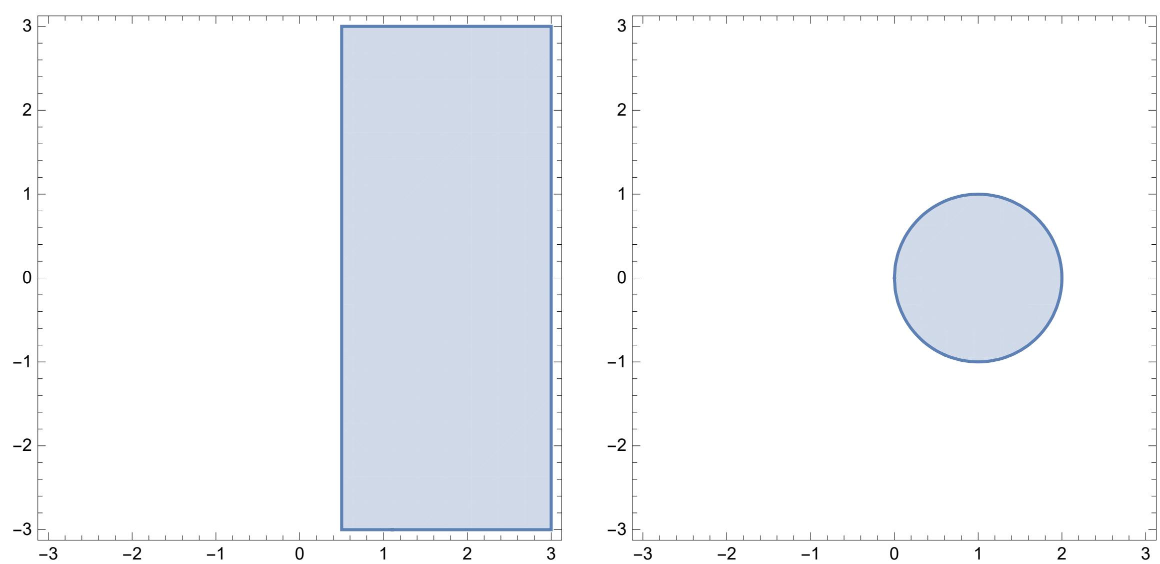
In[1]:=
```
ClearAll
```
Out[1]=
```
ClearAll
```

In[9]:=
```
z = x + ⅈ * y
Solve[w1 == 1 / z1, z1]
```
Out[9]= $x + i\,y$

Out[10]= $\left\{\left\{z1 \to \frac{1}{w1}\right\}\right\}$

In[11]:=
```
A1 = RegionPlot[Re[z] ≥ 1 / 2, {x, -3, 3}, {y, -3, 3}];
A2 = RegionPlot[Re[1 / z] ≥ 1 / 2, {x, -3, 3}, {y, -3, 3}];
GraphicsRow[{A1, A2}]
```

Out[13]=



PRACTICAL-07
25/03/26

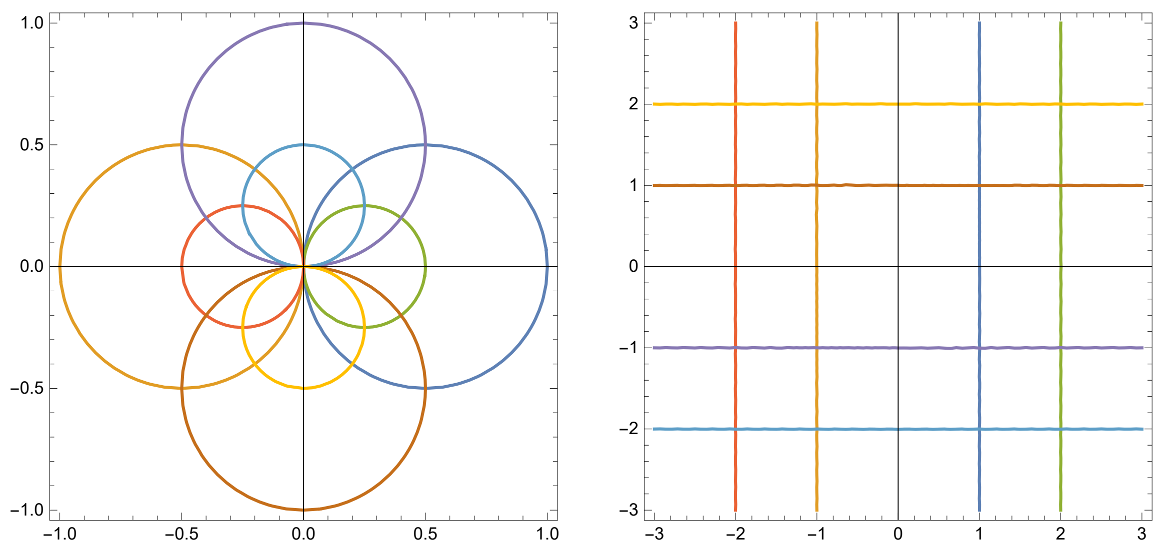
Make a plot of the vertical lines x=a for a=-1,-1/2,1/2,1 and the horizontal lines y=b for b=-1,-1/2,1/2,1. Find the plot of the grid under the mapping w=f(z)=1/z.

In[1]:=
```
ClearAll
```
Out[1]=
```
ClearAll
```

In[2]:=
```
z = x + ⅈ * y
```
Out[2]=
```
x + ⅈ y
```

```
In[6]:= A1 = ContourPlot[{Abs[z - 1/2] == 1/2, Abs[z + 1/2] == 1/2,
      Abs[z - 1/4] == 1/4, Abs[z + 1/4] == 1/4, Abs[z - I/2] == 1/2, Abs[z + I/2] == 1/2,
      Abs[z - I / 4] == 1/4, Abs[z + I / 4] == 1/4}, {x, -1, 1}, {y, -1, 1}, Axes → True];
A2 = ContourPlot[{Abs[(1/z) - 1/2] == 1/2, Abs[(1/z) + 1/2] == 1/2,
      Abs[(1/z) - 1/4] == 1/4, Abs[(1/z) + 1/4] == 1/4,
      Abs[(1/z) - I/2] == 1/2, Abs[(1/z) + I/2] == 1/2,
      Abs[(1/z) - I / 4] == 1/4, Abs[(1/z) + I / 4] == 1/4},
    {x, -3, 3}, {y, -3, 3}, Axes → True];
GraphicsRow[
 {A1,
  A2}]
```

Out[8]=

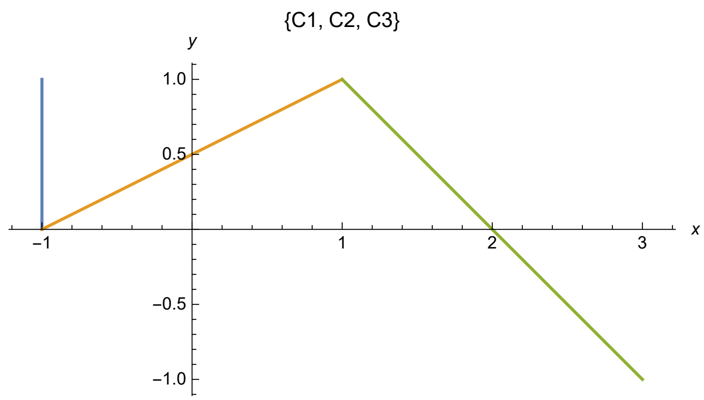


PRACTICAL-08

25/03/26

In[9]:=
```
C1 = {-1, 1 - t};
C2 = {-1 + 2 * t, t};
C3 = {1 + 2 * t, 1 - 2 * t};
ParametricPlot[{C1, C2, C3}, {t, 0, 1}, PlotLabel → {"C1", "C2", "C3"}, AxesLabel → {x, y}]
```

Out[12]=



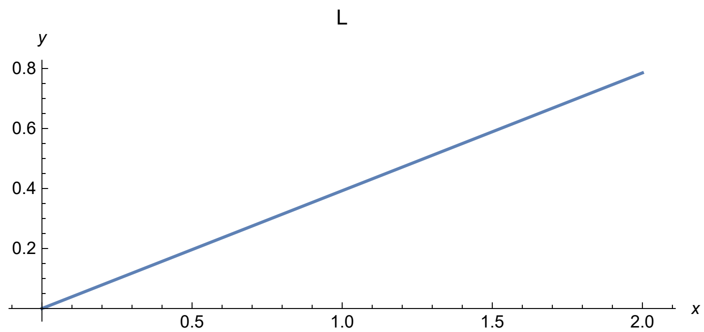
13/4/26

Practical-09

Plot the segment 'L' joining the point A =0 to B=2+i(π/4) and give an exact calculation $\int_L^{upper} e^z \, dz$

```
In[ ]:= ParametricPlot[{2 * t, (Pi / 4) * t}, {t, 0, 1}, AxesLabel → {x, y}, PlotLabel → "L"]
```

Out[ ]=




```
In[ ]:= z = x + i * y
```

Out[ ]= $x + i\,y$

```
In[ ]:= f[z_] := Exp[z]
        g[t_] := (2 + iπ / 4) * t
        ∫_0^1 f[g[t]] * g'[t] dt
```

Out[ ]= $-1 + e^{2+\frac{i\pi}{4}}$

Practical-10

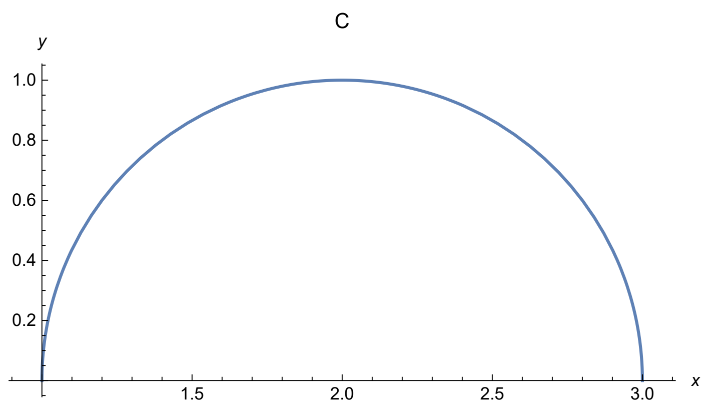
Plot the Semicircle ‘C’ with radius 1 centered at z=2 and evaluate the contour integral

$$\int_C^{upper} 1/(z-2)\, dz$$

```
In[ ]:= ClearAll
Out[ ]= ClearAll
```

```
In[ ]:= ParametricPlot[{2 + Cos[t], Sin[t]}, {t, 0, Pi}, AxesLabel → {x, y}, PlotLabel → "C"]
```

Out[ ]=




```
In[ ]:= f[z_] := 1 / (z - 2)
g[t_] := 2 + Exp[ⅈ * t]
```

$$\int_0^{\pi} f[g[t]] * g'[t]\, dt$$

Out[ ]= $i\,\pi$

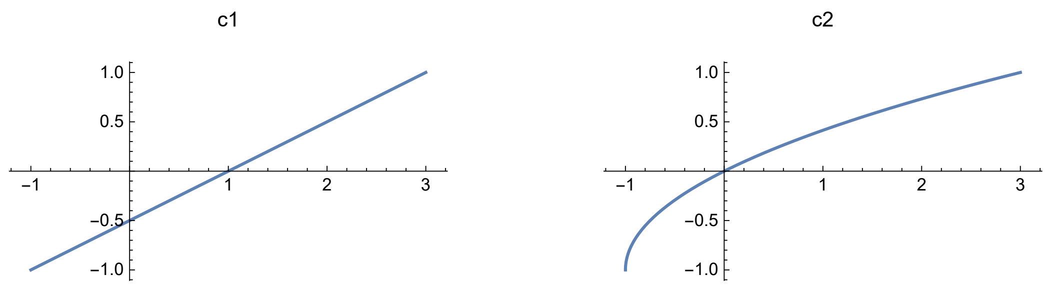
Practical-11

Show that $\int c1z\, dz = \int c2z\, dz = 4+2i$ where c1 is the line-segment from -1-i to 3+i and c2 is the portion of the parabola x=y^2+2y joining -1-i to 3+i. Make plots of two contours c1 and c2

joining -1-i to 3+i.

In[∘]:=
```
c1 = ParametricPlot[{2 * t + 1, t}, {t, -1, 1}, PlotLabel → "c1"];
c2 = ParametricPlot[{t^2 + 2 * t, t}, {t, -1, 1}, PlotLabel → "c2"];
GraphicsRow[{c1, c2}]
```

Out[∘]=



In[∘]:=
```
f[z_] := z
g[t_] := (2 * t + 1) + ⅈ * t
h[t_] := (t^2 + 2 * t) + ⅈ * t
```

$$\int_{-1}^{1} f[g[t]] * g'[t] \, dt$$

$$\int_{-1}^{1} f[h[t]] * h'[t] \, dt$$

Out[∘]= 4 + 2 ⅈ

Out[∘]= 4 + 2 ⅈ

Practical-12

Use ML-inequality to show that $| \int 1/(z^2+1)\, dz| <= 1/2$ sqrt 5 where C is the straight line segment from 2 to 2+i. While solving represent the distance from the point z to the points i and -i respectively i.e; |z+i| and |z-i| on the complex plane C

```
In[1]:= f[z_] := 1/(z^2+1);
        g[t_] := 2 + i * t;
        val = ∫_0^1 f[g[t]] * g'[t] dt;
        ComplexExpand[val]
```

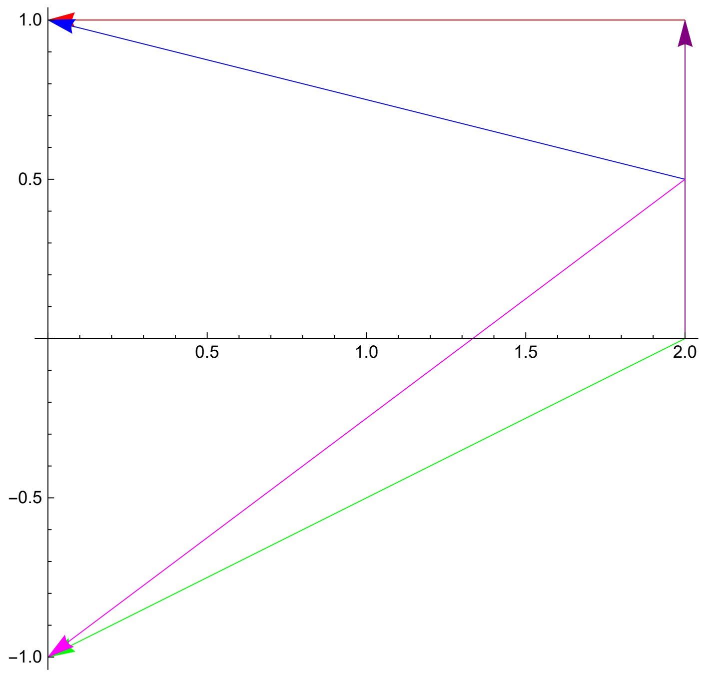
$$\text{Out[4]=}\quad \frac{3\pi}{8} - \text{ArcTan}[2] + i\left(-\frac{\text{Log}[2]}{2} + \frac{\text{Log}[8]}{4}\right)$$

```
In[5]:= N[Abs[ComplexExpand[val]]];
        N[1/2 Sqrt[5]]
```

```
Out[6]= 1.11803
```

In[7]:=
```
Show[Graphics[{Purple, Arrow[{{2, 0}, {2, 1}}], Red,
   Arrow[{{2, 1}, {0, 1}}], Green, Arrow[{{2, 0}, {0, -1}}], Blue,
   Arrow[{{2, 1/2}, {0, 1}}], Magenta, Arrow[{{2, 1/2}, {0, -1}}]}], Axes → True]
```

Out[7]=



17/04/26

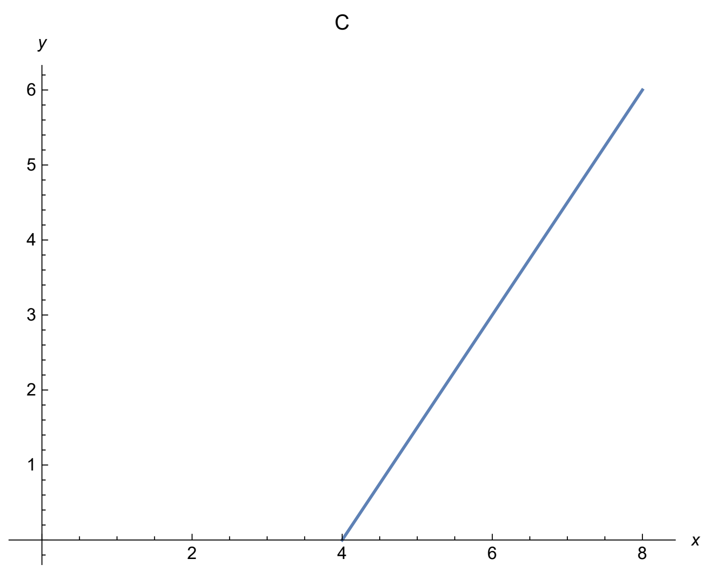
Practical-13

Show that $|\int_C^{\text{upper}} 1/(2\sqrt{z})\,dz$=1+i where $\sqrt{z}$ is the principal branch of the square root function and C is the line segment joining 4 to 8+6i. Also plot the path of integration.

```
In[10]:= f[z_] := 1/(2*z^(1/2))
         g[t_] := 4 + (4 + i*6)*t
         ∫_0^1 f[g[t]]*g'[t] dt
Out[12]= 1 + i

In[13]:= ParametricPlot[{4 + 4*t, 6*t}, {t, 0, 1},
          AxesLabel → {x, y}, AxesOrigin → {0, 0}, PlotLabel → "C"]
```

Out[13]=



Practical-14

Laurentz Series Expansion:

Q.1 Find the Laurentz Series of a given function f(z) around a given ppint z.

1)f(z)=(Sin z-1)/(z^4)

In[20]:=
```
f[z_] := (Sin[z] - 1) / z^4;
L[z_] = Normal[Series[f[z], {z, 0, 14}]]
```

Out[21]=
$$-\frac{1}{z^4}+\frac{1}{z^3}-\frac{1}{6\,z}+\frac{z}{120}-\frac{z^3}{5040}+\frac{z^5}{362\,880}-\frac{z^7}{39\,916\,800}+\frac{z^9}{6\,227\,020\,800}-\frac{z^{11}}{1\,307\,674\,368\,000}+\frac{z^{13}}{355\,687\,428\,096\,000}$$

---

2)f(z)=(Cosz-1)/(z^4)

In[22]:=
```
f[z_] := (Cos[z] - 1) / z^4;
L[z_] = Normal[Series[f[z], {z, 0, 14}]]
```

Out[23]=
$$\frac{1}{24}-\frac{1}{2\,z^2}-\frac{z^2}{720}+\frac{z^4}{40\,320}-\frac{z^6}{3\,628\,800}+\frac{z^8}{479\,001\,600}-\frac{z^{10}}{87\,178\,291\,200}+\frac{z^{12}}{20\,922\,789\,888\,000}-\frac{z^{14}}{6\,402\,373\,705\,728\,000}$$

---

3)f(z)=(Cotz-1)/(z^4)

In[24]:=
```
f[z_] := (Cot[z] - 1) / z^4;
L[z_] = Normal[Series[f[z], {z, 0, 14}]]
```

Out[25]=
$$\frac{1}{z^5}-\frac{1}{z^4}-\frac{1}{3\,z^3}-\frac{1}{45\,z}-\frac{2\,z}{945}-\frac{z^3}{4725}-\frac{2\,z^5}{93\,555}-\frac{1382\,z^7}{638\,512\,875}-\frac{4\,z^9}{18\,243\,225}-\frac{3617\,z^{11}}{162\,820\,783\,125}-\frac{87\,734\,z^{13}}{38\,979\,295\,480\,125}$$

Find and plot three different Laurentz series representation for the function f(z)=3/(2+z-z^2) involving powers of z.

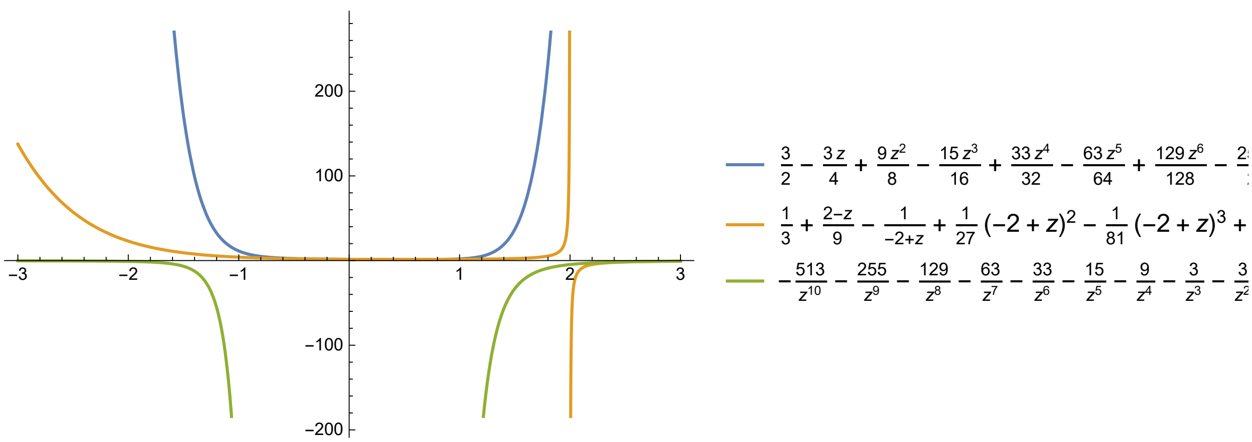
In[26]:=
```
Grid[
 Table[{n, Normal[Series[3 / (2 + z - z^2), {z, n, 10}]]}, {n, {0, 2, Infinity}}], Frame → All]
```

Out[26]=

0	$\frac{3}{2}-\frac{3z}{4}+\frac{9z^2}{8}-\frac{15z^3}{16}+\frac{33z^4}{32}-\frac{63z^5}{64}+\frac{129z^6}{128}-\frac{255z^7}{256}+\frac{513z^8}{512}-\frac{1023z^9}{1024}+\frac{2049z^{10}}{2048}$
2	$\frac{1}{3}+\frac{2-z}{9}-\frac{1}{-2+z}+\frac{1}{27}(-2+z)^2-\frac{1}{81}(-2+z)^3+\frac{1}{243}(-2+z)^4-\frac{1}{729}(-2+z)^5+\frac{(-2+z)^6}{2187}-\frac{(-2+z)^7}{6561}+\frac{(-2+z)^8}{19683}-\frac{(-2+z)^9}{59049}+\frac{(-2+z)^{10}}{177147}$
∞	$-\frac{513}{z^{10}}-\frac{255}{z^9}-\frac{129}{z^8}-\frac{63}{z^7}-\frac{33}{z^6}-\frac{15}{z^5}-\frac{9}{z^4}-\frac{3}{z^3}-\frac{3}{z^2}$

In[27]:=
```
Plot[Evaluate[Table[{Normal[Series[3 / (2 + z - z^2), {z, n, 10}]]}, {n, {0, 2, Infinity}}]],
 {z, -3, 3}, PlotLegends → "Expressions"]
```

Out[27]=



Practical-15

Locate the poles of f(z)=1/(5z^4+26z^2+5) and specify their order

In[43]:=
```
f[z_] := 1 / (5 z^4 + 26 z^2 + 5);
Print["f[z]=", f[z]];
Print [Denominator[f[z]], "=0"];
solset = Sort[Solve[Denominator[f[z]] == 0, z]];
Print[solset]
```

$$f[z]=\frac{1}{5+26z^2+5z^4}$$

$$5+26z^2+5z^4=0$$

$$\left\{\left\{z\to-\frac{i}{\sqrt{5}}\right\},\left\{z\to\frac{i}{\sqrt{5}}\right\},\left\{z\to-i\sqrt{5}\right\},\left\{z\to i\sqrt{5}\right\}\right\}$$

Practical-16

Locate the zeros and poles of g(z)=πCot(nπ)/z^2 and determine their order. Also justify that Res(g,0)=-π^2/3

In[50]:=
```
g[z_] := (Pi * Cot[Pi z]) / z^2
Reduce[g[z] == 0, z]
```

Out[51]= $c_1 \in \mathbb{Z}$ && $z \neq 0$ && $z == \frac{\frac{\pi}{2}+\pi c_1}{\pi}$

In[52]:=
```
Poles = Solve[Denominator[g[z]] == 0, z]
```

Out[52]= {{z → 0}, {z → 0}}

In[53]:=
```
residue = Residue[g[z], {z, 0}]
```

Out[53]= $-\frac{\pi^2}{3}$

Practical-17

Evaluate $\int \mathrm{Exp}(2/z)\,dz$, where C(0) denotes the circle {z:|z|=1} with positive orientation

In[58]:=
```
ClearAll;
f[z_] := Exp[2 / z]
g[t_] := Cos[t] + i * Sin[t]
Integrate[f[g[t]] * g'[t], {t, 0, 2 Pi}]
```

Out[61]= $4\,i\,\pi$